Module 3: Resampling and Simulation-Based Statistical Inference

MA 563 – Statistical Computation and Simulation

Dr. Rishikesh Yadav

rishikesh@iitmandi.ac.in — TA: Vedant Vibhor (D24228@students.iitmandi.ac.in)
School of Mathematics and Statistical Sciences, IIT Mandi

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Module 3 at a Glance

Module 3: Resampling and Simulation-Based Statistical Inference (10 Lectures)

Bootstrap principles, parametric vs nonparametric bootstrap, cross-validation, jackknife (brief), simulation-based uncertainty quantification, computational trade-offs, likelihood-based inference via simulation, parametric simulation for inference, confidence intervals via simulation, hypothesis testing using Monte Carlo, permutation tests, bootstrap-based inference (deep dive), simulation vs asymptotic inference.

How Module 3 builds on Modules 1 and 2

- **Module 1** gave us generators: we can draw from any F we can write down.
- **Module 2** gave us Monte Carlo: any expectation can be approximated by an average.
- **Module 3** applies both to *statistical inference itself*: standard errors, confidence intervals, p-values and prediction error – computed by simulation instead of by formula.

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Every Estimate Needs an Error Bar

- We observe data $x_1, \ldots, x_n \overset{\text{iid}}{\sim} F$ and compute a statistic $\hat{\theta} = s(x_1, \ldots, x_n)$ to estimate a parameter $\theta = T(F)$.
- A point estimate alone is never enough. We also need:
  - its **standard error** $\text{se}_F(\hat{\theta}) = \sqrt{\text{Var}_F(\hat{\theta})}$,
  - its **bias** $\mathbb{E}_F[\hat{\theta}] - \theta$,
  - a **confidence interval** for θ, and/or a p-**value** for a hypothesis about θ.
- All four are properties of the **sampling distribution** of $\hat{\theta}$ – the distribution of $\hat{\theta}$ over hypothetical repetitions of the experiment.

The central question of Module 3

The sampling distribution depends on the *unknown* F, and we see only *one* sample. Can we approximate it anyway – for *any* statistic, without deriving a formula?

The Textbook Case – and Why It Is Special

- For the sample mean $\bar{X}$, classical theory delivers everything:

$$\mathrm{se}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}, \qquad \bar{X} \pm t_{n-1,\,\alpha/2}\,\frac{s}{\sqrt{n}}.$$

- This works because $\bar{X}$ is **linear** in the data and, for Normal data, its distribution is known *exactly*; otherwise the CLT gives a good approximation.
- Change the statistic, and the comfort disappears:
  - **Median:** se $\approx 1/\{2f(m)\sqrt{n}\}$ – needs the unknown density f at the unknown median m.
  - **Correlation r:** the classical formula assumes bivariate Normality.
  - **Ratio, trimmed mean, eigenvalue, a classifier's error rate:** typically no usable formula at all.

The honest picture

Closed-form standard errors exist for a small, special family of statistics. Real analyses use statistics that were chosen for their *scientific* meaning, not for having a tidy variance formula.

Real Problems with No Standard-Error Formula

Medicine: median survival

A small trial ($n = 25$) reports *median* time to recovery. Patients are expensive; the data are skewed. How uncertain is the median?

Finance: the Sharpe ratio

Sharpe ratio $= \bar{r}/s_r$ of a fund's monthly returns – a *ratio* of two random quantities, from heavy-tailed returns. What is its standard error?

Climate: a 100-year rainfall level

From 40 years of annual-maximum rainfall in the Himalaya, estimate the level exceeded once per century – a non-linear function of fitted extreme-value parameters. How wide is its interval?

Machine learning: test error

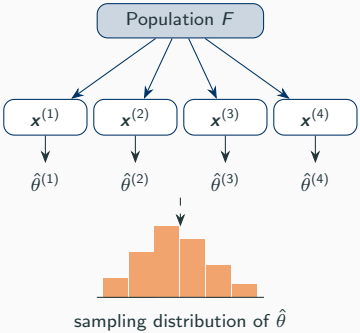
A classifier trained on 500 patients. What error rate will it have on *new* patients – and which of ten candidate models generalizes best?

The common thread

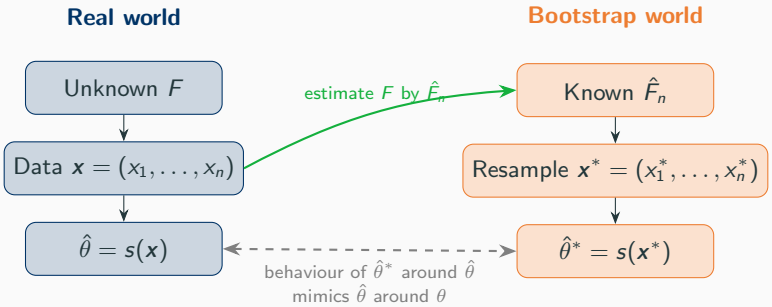
Each asks for a sampling distribution that classical theory cannot supply cleanly. Each is answered in practice by **resampling** or **simulation**.

What We Would Do If We Could Repeat the Experiment

- If we could draw *fresh* samples from F again and again, we would simply compute $\hat{\theta}$ on each and look at the histogram.
- This is exactly Monte Carlo (Module 2), with $h(\cdot)$ replaced by the whole estimation procedure.
- The obstacle: F is unknown, and nature lets us draw from it only **once**.



The Resampling Idea: Let the Sample Stand In for the Population



Efron (1979): the bootstrap

Everything we cannot compute in the real world (because F is unknown) we **can** compute in the bootstrap world (because $\hat{F}_n$ is known) – by simulation.

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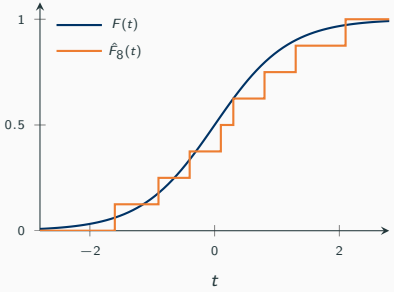
The Empirical Distribution Function

Definition

The empirical distribution $\hat{F}_n$ puts probability mass $1/n$ on each observation:

$$\hat{F}_n(t) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{x_i \le t\}.$$

- **Glivenko–Cantelli:** $\sup_t |\hat{F}_n(t) - F(t)| \xrightarrow{\text{a.s.}} 0$.
- This is the same D_n used in the Kolmogorov–Smirnov test of Module 1.
- $\hat{F}_n$ is the nonparametric MLE of F: the best guess of F when we assume nothing.



The Plug-in Principle

- Many parameters are *functionals* of F: $\theta = T(F)$. The **plug-in estimator** is $\hat{\theta} = T(\hat{F}_n)$.

Parameter	$T(F)$	Plug-in $T(\hat{F}_n)$
Mean	$\int x\, dF(x)$	$\bar{x}$
Variance	$\int (x-\mu)^2 dF(x)$	$\hat{\sigma}^2 = \frac{1}{n}\sum (x_i - \bar{x})^2$ (divisor $n!$)
Median	$F^{-1}(1/2)$	sample median
Correlation	$\mathrm{Cov}(X,Y)/(\sigma_X \sigma_Y)$	sample correlation r

The bootstrap is the plug-in principle applied one level up

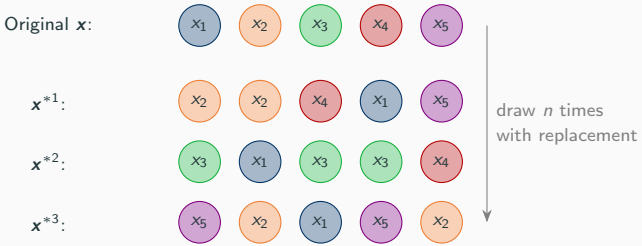
The standard error is *also* a functional of F: $\mathrm{se}_F(\hat{\theta})$. Its plug-in estimate is

$$\mathrm{se}_{\hat{F}_n}(\hat{\theta}^*) = \text{standard deviation of } s(\boldsymbol{x}^*) \text{ when } \boldsymbol{x}^* \overset{\text{iid}}{\sim} \hat{F}_n.$$

This is the **ideal bootstrap estimate** of the standard error.

Sampling from $\hat{F}_n$ = Resampling With Replacement

- $\hat{F}_n$ is a **discrete** distribution with $P(X^* = x_i) = 1/n$, $i = 1, \ldots, n$.
- By the discrete inverse-transform method of Module 1: draw $U \sim \text{Unif}(0, 1)$, set $I = \lceil nU \rceil$, and return x_I.
- Repeating n times gives a bootstrap sample $\boldsymbol{x}^* = (x_{I_1}, \ldots, x_{I_n})$ – some observations appear several times, some not at all.



The Nonparametric Bootstrap: Algorithm

Algorithm: Nonparametric Bootstrap

Given data $\boldsymbol{x} = (x_1, \ldots, x_n)$ and a statistic $s(\cdot)$, with $\hat{\theta} = s(\boldsymbol{x})$:

1. For $b = 1, \ldots, B$:
   (a) Draw $\boldsymbol{x}^{*b} = (x_1^{*b}, \ldots, x_n^{*b})$ by sampling n values **with replacement** from $\boldsymbol{x}$.
   (b) Compute the bootstrap replication $\hat{\theta}_b^* = s(\boldsymbol{x}^{*b})$.
2. Use the empirical distribution of $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$ as an approximation to the sampling distribution of $\hat{\theta}$.

- The resample has the **same size** n as the data – the variability of $\hat{\theta}$ depends on n.
- $s(\cdot)$ is treated as a black box: median, trimmed mean, regression slope, a whole fitted model – the algorithm does not change.
- Step 2 is precisely **crude Monte Carlo** (Module 2) applied to the bootstrap world.

A Toy Example by Hand: $x = (2, 5, 9)$

- $\bar{x} = 16/3 \approx 5.333$. There are $3^3 = 27$ equally likely ordered resamples, but only $\binom{5}{3} = 10$ distinct ones (as multisets), e.g. $(2,2,2)$, $(2,5,9)$, $(9,9,5)$, ...
- Averaging $\bar{x}^*$ over all 27 resamples gives the **exact** (ideal) bootstrap distribution – no simulation needed for such a tiny n.

Exact bootstrap standard error

$$\mathrm{se}_{\hat{F}_n}(\bar{x}^*) = \frac{\hat{\sigma}}{\sqrt{n}} = \sqrt{\frac{8.222}{3}} \approx 1.656,$$

where $\hat{\sigma}^2 = \frac{1}{n}\sum(x_i - \bar{x})^2 = 8.222$.

Compare with the textbook

$s/\sqrt{n} = \sqrt{12.333/3} \approx 2.028.$

$$\frac{\mathrm{se}_{\mathrm{boot}}}{s/\sqrt{n}} = \sqrt{\frac{n-1}{n}} = 0.816.$$

Lesson: the bootstrap uses plug-in (divisor-n) variability, so it slightly *underestimates* spread for very small n – the factor $\sqrt{(n-1)/n} \to 1$ quickly.

How Big Is the Bootstrap World?

- The number of distinct bootstrap samples of size n is $\binom{2n-1}{n}$:

$n = 3$	$n = 10$	$n = 20$
10	92,378	$\approx 6.9 \times 10^{10}$

- Exact enumeration is hopeless beyond tiny n – so we approximate the ideal bootstrap by drawing B resamples at random: **Monte Carlo inside the bootstrap world**.

Two separate layers of approximation

$$\underbrace{\mathrm{se}_F(\hat{\theta})}_{\text{truth}} \overset{\text{statistical error}}{\approx} \underbrace{\mathrm{se}_{\hat{F}_n}(\hat{\theta}^*)}_{\text{ideal bootstrap}} \overset{\text{Monte Carlo error}}{\approx} \underbrace{\widehat{\mathrm{se}}_B}_{\text{what we compute}}$$

Increasing B removes only the second error. The first is fixed by the data (it shrinks only with n).

Bootstrap Standard Error and Bias

Bootstrap estimates from B replications

With $\bar{\theta}^* = \frac{1}{B}\sum_{b=1}^{B} \hat{\theta}_b^*$,

$$\widehat{\mathrm{se}}_B = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\theta}_b^* - \bar{\theta}^*\right)^2}, \qquad \widehat{\mathrm{bias}}_B = \bar{\theta}^* - \hat{\theta}.$$

- **Why the bias formula?** In the real world, bias $= \mathbb{E}_F[\hat{\theta}] - \theta$. Plug in: $F \to \hat{F}_n$, $\theta \to \hat{\theta}$, $\hat{\theta} \to \hat{\theta}^*$.
- **Bias-corrected estimator:** $\tilde{\theta} = \hat{\theta} - \widehat{\mathrm{bias}}_B = 2\hat{\theta} - \bar{\theta}^*$.

Use bias correction with care

Bias correction can *increase* variance. A common rule: correct only if $|\widehat{\mathrm{bias}}_B| / \widehat{\mathrm{se}}_B > 0.25$; otherwise report $\hat{\theta}$ and note the bias is negligible.

Worked Example: Hospital Length of Stay

A hospital records length of stay (days) for $n = 20$ patients – small sample, right-skewed:

1.2, 1.3, 1.7, 2.0, 2.2, 2.2, 2.7, 2.9, 3.2, 3.2, 3.6, 3.8, 3.9, 3.9, 4.6, 5.7, 7.3, 9.0, 9.3, 9.3

Classical

$\bar{x} = 4.15, \quad s = 2.617$

$s/\sqrt{n} = 0.585$

Bootstrap ($B = 4999$)

$\widehat{\text{se}}_B = 0.571 \quad (\approx \sqrt{19/20} \times 0.585)$

$\widehat{\text{bias}}_B = 0.002 \quad$ (negligible, as theory says)

- For the mean, the bootstrap simply reproduces theory – a useful **sanity check** of your code.
- The payoff comes when you switch $s(\cdot)$ to the *median* or a 10% *trimmed mean*: the same five lines of code give a standard error that classical theory cannot.

We will run this example (and the median version) in the lab.

Why Does the Bootstrap Work? (Sketch)

- The bootstrap approximates the law of a *root*, e.g. $R = \sqrt{n}(\hat{\theta} - \theta)$ under F, by the law of $R^* = \sqrt{n}(\hat{\theta}^* - \hat{\theta})$ under $\hat{F}_n$.

Theorem (Bootstrap consistency for the mean; Bickel & Freedman, 1981)

If $X_1, \ldots, X_n \overset{\text{iid}}{\sim} F$ with $\mathbb{E}X^2 < \infty$, then almost surely

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}_*\big(\sqrt{n}(\bar{X}^* - \bar{X}) \le x\big) - \mathbb{P}_F\big(\sqrt{n}(\bar{X} - \mu) \le x\big) \right| \longrightarrow 0.$$

The result extends to smooth functions of means (delta method), and many other smooth statistics.

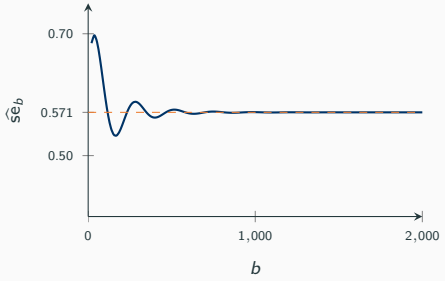
- **Intuition:** the law of a good root depends on F only through a few features (variance, skewness, ...), and $\hat{F}_n$ gets those features right as n grows.
- The theorem needs **smoothness** and **finite variance** – we will see what happens when they fail (Section 9).

How Many Bootstrap Resamples B?

Rules of thumb (Efron & Tibshirani, 1994)

- Standard error: $B = 200$ is usually plenty.
- Percentile-based confidence intervals: $B \geq 1000$–2000 (tails need more draws).
- p-values: $B = 999$ or 9999 (see Section 7).

- Monte Carlo error of anything bootstrap-based shrinks like $1/\sqrt{B}$ – the Module 2 rate.
- Use the Module 2 **running-mean trace plot** on $\widehat{\mathrm{se}}_b$ against b: stop once it is flat.



trace settles near the ideal value

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The Parametric Bootstrap

- If we trust a parametric model $\{F_\theta\}$, estimate F by $F_{\hat{\theta}}$ instead of $\hat{F}_n$.
- We then simulate **new** data from the fitted model – using the generators of Module 1.

Algorithm: Parametric Bootstrap

1. Fit the model: $\hat{\theta}$ (typically the MLE) from $\boldsymbol{x}$.
2. For $b = 1, \ldots, B$:
   (a) Simulate $x_1^{*b}, \ldots, x_n^{*b} \overset{\text{iid}}{\sim} F_{\hat{\theta}}$.
   (b) Re-fit the model to $\boldsymbol{x}^{*b}$; record $\hat{\theta}_b^*$ (or any statistic of interest).
3. Use $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$ exactly as in the nonparametric case.

Key difference

Nonparametric bootstrap: *resamples the observed values*. Parametric bootstrap: *generates new values* from a fitted model – so it can produce values never seen in the data (e.g. more extreme ones).

Example: Exponential Lifetimes (Reliability Engineering)

- Lifetimes of n components are modelled as $X_i \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$. Engineers report the **median lifetime** $m = \ln 2/\lambda$.
- MLE: $\hat{\lambda} = 1/\bar{x}$, so $\hat{m} = \bar{x} \ln 2$.

Parametric bootstrap

1. Simulate $X_i^* = -\ln(U_i)/\hat{\lambda}, \quad U_i \overset{\text{iid}}{\sim} \text{Unif}(0,1)$ (inverse transform, Module 1).
2. Compute $\hat{m}^* = \bar{X}^* \ln 2$; repeat B times; take the standard deviation.

Check against exact theory

Under $F_{\hat{\lambda}}$, $\bar{X}^* \sim \text{Gamma}(n,\ \text{rate } n\hat{\lambda})$, so $\text{se}(\hat{m}^*) = \bar{x} \ln 2/\sqrt{n}$ exactly. The parametric bootstrap reproduces the exact answer when one exists – and keeps working (for the 10th percentile, a ratio of lifetimes, ...) when it does not.

Nonparametric vs Parametric: Side by Side

	Nonparametric	**Parametric**
Estimate of F	$\hat{F}_n$ (empirical)	$F_{\hat{\theta}}$ (fitted model)
Simulation step	resample observed x_i with replacement	draw fresh values from $F_{\hat{\theta}}$
Main assumption	data are i.i.d. and representative	the model family is correct
Strength	robust to model misspecification	more accurate when model is right; works for small n and tails
Main risk	poor for very small n; cannot go beyond observed range	silently wrong if model is wrong

A useful habit

Run **both**. If they agree, report with confidence. If they disagree, the disagreement is itself a diagnostic – often of model misspecification.

Bootstrapping Regression Models

Model: $y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$, fitted by least squares: $\hat{\boldsymbol{\beta}}$, residuals $e_i = y_i - \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}$.

Pairs (case) bootstrap

1. Resample pairs $(\boldsymbol{x}_i, y_i)$ with replacement.
2. Refit to get $\hat{\boldsymbol{\beta}}^*$.

+ valid under heteroscedasticity.
− design changes across resamples.

Residual bootstrap

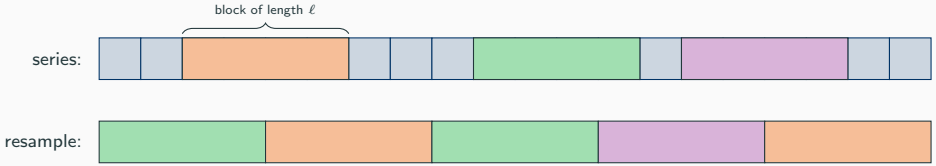
1. Resample residuals $e_1^*, \ldots, e_n^*$ from $\{e_i\}$.
2. Set $y_i^* = \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}} + e_i^*$; refit.

+ keeps the design fixed (designed experiments).
− assumes i.i.d. errors.

- The residual bootstrap is **semi-parametric**: the mean structure is a model, the error distribution is empirical.
- Medicine example: dose–response regression from a clinical study – pairs bootstrap is the safe default because variability usually grows with dose.

Dependent Data: The Moving Block Bootstrap

- Resampling individual observations **destroys** autocorrelation: a resampled daily rainfall series looks like white noise, so standard errors come out far too small.
- **Fix (Künsch, 1989):** resample *blocks* of ℓ consecutive observations and glue them together.



- Within each block the dependence is preserved; it is broken only at the n/ℓ joins.
- Block length trades bias against variance; a common rate is $\ell \propto n^{1/3}$.
- **Climate:** trends in monsoon rainfall need block (e.g. by season) resampling – an i.i.d. bootstrap gives overconfident intervals.

Which Bootstrap When?

Data situation	Bootstrap	Resampling unit
i.i.d., no trusted model	Nonparametric	observations x_i
i.i.d., trusted parametric model, small n	Parametric	fresh draws from $F_{\hat{\theta}}$
Regression, random design / heteroscedastic	Pairs	cases $(\boldsymbol{x}_i, y_i)$
Regression, fixed design, i.i.d. errors	Residual	residuals e_i
Time series / spatial data	Block	blocks of neighbours
Clustered data (patients, schools)	Cluster	whole clusters

The golden rule

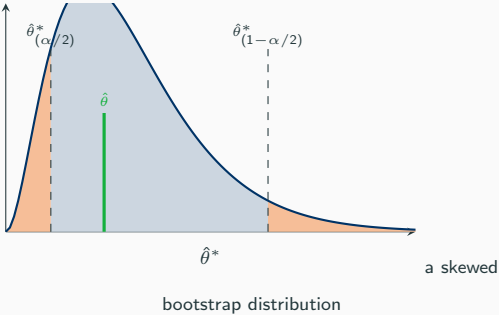
Resample the way the data were generated. The bootstrap replicates whatever randomness you build into the resampling step – and nothing else.

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From a Bootstrap Distribution to an Interval

- After B resamples we hold a histogram of $\hat{\theta}^*_1, \ldots, \hat{\theta}^*_B$.
- Write $\hat{\theta}^*_{(\alpha)}$ for its α-quantile (the $(B\alpha)$-th ordered value).
- There are several ways to turn this histogram into a $(1-\alpha)$ interval; they differ in **what they assume** and **how accurate** they are.



The Normal and Percentile Intervals

1. Normal (standard) bootstrap interval

$$\hat{\theta} \pm z_{\alpha/2}\, \widehat{\mathrm{se}}_B.$$

Uses the bootstrap only for the standard error; assumes $\hat{\theta}$ is approximately Normal and unbiased.

2. Percentile interval

$$\left[\ \hat{\theta}^*_{(\alpha/2)},\quad \hat{\theta}^*_{(1-\alpha/2)}\ \right].$$

Read the interval directly off the bootstrap histogram.

- The percentile interval is **transformation-respecting**: if $[L, U]$ is the interval for θ, then $[g(L), g(U)]$ is the interval for $g(\theta)$ for monotone g (e.g. log, odds).
- It is exactly right if some (unknown) transformation makes $\hat{\theta}$ Normal and unbiased – but it does not correct for bias or skewness on its own.

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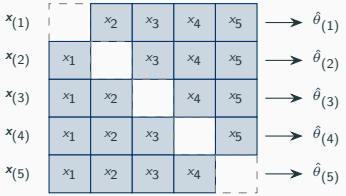
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The Jackknife: Leave One Out

- Instead of random resamples, use the n **deterministic** leave-one-out samples

$$\boldsymbol{x}_{(i)} = (x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n).$$

- Compute $\hat{\theta}_{(i)} = s(\boldsymbol{x}_{(i)})$, $i = 1, \ldots, n$, and their mean $\hat{\theta}_{(\cdot)} = \frac{1}{n} \sum_i \hat{\theta}_{(i)}$.



Jackknife Bias and Standard Error

Jackknife estimates

$$\widehat{\text{bias}}_{\text{jack}} = (n-1)\big(\hat{\theta}_{(\cdot)} - \hat{\theta}\big), \qquad \widehat{\text{se}}_{\text{jack}} = \sqrt{\frac{n-1}{n}\sum_{i=1}^{n}\big(\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)}\big)^2},$$

and the bias-corrected estimator $\hat{\theta}_{\text{jack}} = n\hat{\theta} - (n-1)\hat{\theta}_{(\cdot)}$.

- The inflation factors $(n-1)$ compensate for the leave-one-out samples being *very* close to the original.

Two checks you can do by hand

- **Sample mean:** $\hat{\theta}_{(i)} = (n\bar{x} - x_i)/(n-1)$, which gives $\widehat{\text{se}}_{\text{jack}} = s/\sqrt{n}$ *exactly*.
- **Plug-in variance** $\hat{\sigma}^2$ (divisor n): the jackknife bias correction returns s^2 (divisor $n-1$) *exactly* – it removes the $O(1/n)$ bias.

Check 1 (Mean): why $\widehat{\mathrm{se}}_{\mathrm{jack}} = s/\sqrt{n}$

Let $\hat{\theta} = \bar{x}$ and $\hat{\theta}_{(i)} = \bar{x}_{(i)}$.

$$\bar{x}_{(i)} = \frac{1}{n-1} \sum_{j \neq i} x_j = \frac{n\bar{x} - x_i}{n-1}.$$

Average the leave-one-out means:

$$\hat{\theta}_{(\cdot)} = \frac{1}{n} \sum_{i=1}^{n} \bar{x}_{(i)} = \bar{x}.$$

Hence

$$\bar{x}_{(i)} - \bar{x} = \frac{\bar{x} - x_i}{n-1} \quad \Rightarrow \quad \sum_{i=1}^{n} (\bar{x}_{(i)} - \bar{x})^2 = \frac{1}{(n-1)^2} \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

Plugging into the jackknife SE formula gives

$$\widehat{\mathrm{se}}^2_{\mathrm{jack}} = \frac{n-1}{n} \cdot \frac{1}{(n-1)^2} \sum_{i=1}^{n} (x_i - \bar{x})^2 = \frac{1}{n(n-1)} \sum_{i=1}^{n} (x_i - \bar{x})^2 = \frac{s^2}{n}.$$

Where the Jackknife Fails – and Bootstrap vs Jackknife

Failure for non-smooth statistics

For the **sample median**, deleting one point moves $\hat{\theta}_{(i)}$ to one of only two values, so $\widehat{\text{se}}_{\text{jack}}$ is *inconsistent* – it does not converge to the true standard error. (The delete-d jackknife with $d \to \infty$ repairs this.)

	Jackknife	Bootstrap
Samples	n deterministic leave-one-out	B random resamples
Randomness in answer	none (reproducible)	Monte Carlo error
Smooth statistics	good (a linear approximation)	good
Median, quantiles	fails	works
Confidence intervals	not directly	yes

- Today the jackknife's main roles: cheap bias correction.

Outline

1. Motivation: Inference When Formulas Run Out
2. The Bootstrap Principle
3. Parametric vs Nonparametric Bootstrap
4. Bootstrap Confidence Intervals
5. The Jackknife (Brief)
6. Monte Carlo Hypothesis Testing and Permutation Tests
   - Monte Carlo p-Values
   - Parametric Bootstrap Tests and Simulated Power
   - Permutation Tests
   - Example: A Two-Sample Permutation Test
   - Exchangeability: Permutation vs Bootstrap Tests
7. Unit Summary

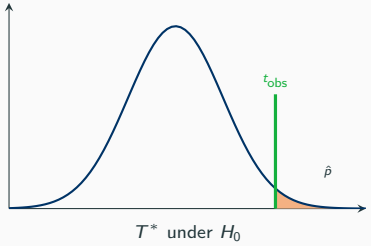
Testing Without a Null Distribution Table

- Classical testing needs the **null distribution** of the statistic T: t, χ^2, F tables.
- For a new or complicated statistic, that distribution is unknown – but if we can **simulate data under** H_0, we can simulate T under H_0.

Algorithm: Monte Carlo test

1. Compute $t_{\text{obs}} = T(\boldsymbol{x})$.
2. For $b = 1, \ldots, B$: simulate $\boldsymbol{x}^{*b}$ under H_0; compute T_b^*.
3. Return

$$\hat{p} = \frac{1 + \#\{b : T_b^* \geq t_{\text{obs}}\}}{B + 1}.$$



Simulating Size and Power

- Monte Carlo also tells us how a test **behaves**, before we ever see real data.

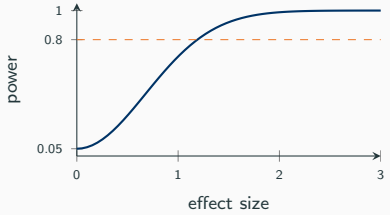
Algorithm: Simulated rejection rate

For a data-generating distribution G:

1. Repeat N times: simulate a dataset from G, run the test, record whether it rejects at level α.
2. Rejection rate $\hat{\pi}(G) = (\#\text{rejections})/N$, with se $\approx \sqrt{\hat{\pi}(1-\hat{\pi})/N}$.

If G satisfies H_0, $\hat{\pi}$ estimates the **size**; otherwise it estimates the **power**.

- **Clinical trials:** sample-size planning when the outcome model is too complex for a power formula.
- **A/B testing:** how many users before a 1% uplift is detectable with 80% power?



Computational Trade-offs

Procedure	Number of model fits	Remark
Bootstrap SE	$B \approx 200$	cheap
Percentile	$B \approx 2000$	moderate
Jackknife	n	deterministic

- Resampling loops are **embarrassingly parallel**: each replicate is independent – spread across cores.
- Use parallel-safe, **seeded** streams (Module 1) so results are reproducible.
- Spend the budget where error matters: increasing B beyond the point where Monte Carlo error is small compared with the statistical error buys nothing.

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Putting It All Together

Method	Core idea	Best suited for
Nonparametric bootstrap	Resample data with replacement	SE, bias, CI of any statistic
Parametric bootstrap	Simulate from fitted model	Small n, tails, MLE-based inference
Jackknife	Leave one out	Bias correction, BCa acceleration
Monte Carlo test	Simulate T under H_0	Non-standard statistics, composite H_0